YINGDAN SHI

yshi73@hawk.illinoistech.edu ◇ shi-d.github.io ◇ Google Scholar ◇ LinkedIn

EDUCATION

Illinois Institute of Technology — **Aug. 2024 - Jun. 2028**
Ph.D. in Computer Engineering. Advisor: Prof. Ren Wang — GPA: 4.0/4.0

Soochow University — **Sep. 2021 - Jun. 2024**
M.E. in Computer Technology. Advisor: Prof. Jingya Zhou — GPA: 3.8/4.0

Hangzhou Normal University — **Sep. 2017 - Jun. 2021**
B.E. in Computer Science and Technology. — GPA: 3.8/4.0

RESEARCH INTERESTS

Trustworthy post-training for LLMs and LLM-based agents, with a focus on removing, editing, and controlling learned knowledge and behaviors.
Keywords: Post-training, Safety Alignment, Agent Safety, Self-evolving Agents, Unlearning

RESEARCH EXPERIENCE

TIML Lab, Illinois Institute of Technology — Aug. 2024 – Present

LLM Agents & Post-training

- **Trajectory Unlearning for LLM Agents** [Under review, first author]. Formulated *trajectory unlearning*, extending unlearning from factual knowledge to **long-horizon** agent *behaviors*. Built agent unlearning benchmarks on **ALFWorld** and **WebShop**, and proposed GiRPO, a GRPO-based on-policy RL algorithm that reduces similarity to forgotten trajectories to **3%** on Qwen2.5-7B-Instruct while preserving task success rate.
- **Self-Unlearning for Self-Evolving Agents** [Ongoing]. Found that **self-evolving agents** silently re-acquire unlearned knowledge while evolving; designing self-unlearning inside the evolution loop to prevent re-memorization.
- **Knowledge Entanglement in LLMs** [Ongoing]. Modeled knowledge as hidden-state *subspaces* (not single vectors) and quantified inter-knowledge entanglement to guide targeted, low-collateral unlearning in LLMs.

Machine Unlearning & Trustworthy ML

- **Plasticity Collapse in Continual Unlearning** [ECCV 2026, first author]. Identified *plasticity collapse*: models gradually fail to forget new requests and re-memorize previously forgotten data. Traced it theoretically to accumulating geometric constraints in parameter space; validated on classification, DDPM, and LLaVA.
- **Nonlinear Unlearning via Mode Connectivity** [ICML 2026, first author]. Showed linear unlearning updates suffer from weight entanglement; proposed a nonlinear pathway along mode-connected curves, a plug-and-play add-on to existing methods that mitigates over- and under-forgetting.
- **Fake Forgetting** [ICML 2026, first author]. Used conformal prediction to show samples "forgotten" under accuracy metrics remain exposed; proposed an uncertainty-aware metric and unlearning framework to close the gap.
- **GNN Watermarking** [ICML 2026, co-first author]. Exposed ownership ambiguity and poisoning risks of backdoor watermarks; proposed an explanation-based watermark requiring no training-data manipulation.

SELECTED PUBLICATIONS

(* co-first author)

[1] **Yingdan Shi**, Xiang Xu, Kaize Ding, Alfred O. Hero and Ren Wang. "On the Plasticity Collapse in Continual Machine Unlearning." **ECCV 2026**. [Paper][Code]

[2] **Yingdan Shi** and Ren Wang. "Exploring Nonlinear Pathway in Parameter Space for Machine Unlearning." **ICML 2026**. [Paper][Code]

[3] **Yingdan Shi**, Sijia Liu, Kaize Ding and Ren Wang. "Tackling Fake Forgetting through Uncertainty Quantification." **ICML 2026**. [Paper][Code]

[4] Jane Downer*, **Yingdan Shi***, Ziyan Liu*, Ren Wang and Binghui Wang. "Watermarking Graph Neural Networks via Explanations for Ownership Protection." **ICML 2026**. [Paper][Code]

[5] **Yingdan Shi** and Ren Wang. "Trajectory Unlearning on LLM-based Agents." Under review, 2026. [Paper][Code]

[6] Hyoseo Kim, **Yingdan Shi** and Ren Wang. "Unmasking Intent Blindness in Generative Engine Optimization." Under review, 2026.

TECHNICAL SKILLS

Languages: Python, C/C++, SQL, Bash
ML / LLM Frameworks: PyTorch, Hugging Face (Transformers, PEFT, TRL), verl, vLLM, DeepSpeed, PyTorch Geometric
Post-training: SFT, LoRA, DPO, GRPO, on-policy distillation, agentic RL
Infrastructure: Linux, Slurm, Docker, Git, Weights & Biases